10th Grade Midterm Analysis of Decisions: C4 Solutions

Evaluated Criterion

C4: Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case outcome, interpreting risks or limitations.

1 Maximax and Maximin Criteria

When probabilities are not available, a decision-maker can compare alternatives using the payoffs associated with each possible state of nature.

Two important decision criteria are:

- **Maximax**: an optimistic criterion that focuses on the best possible payoff for each alternative.
- **Maximin**: a conservative criterion that focuses on the worst possible payoff for each alternative.

For a payoff table with alternatives in rows and states of nature in columns, let

$$P_{ay}(A_i, S_j)$$

represent the payoff obtained when alternative A_i is selected and state of nature S_j occurs.

For each alternative, calculate

$$M_i = \max_j P_{ay}(A_i, S_j) = \max\{P_{ay}(A_i, S_1), P_{ay}(A_i, S_2), \ldots, P_{ay}(A_i, S_n)\} \quad \longrightarrow \quad A^*_{\text{Maximax}} = \arg\max_i M_i$$

for the Maximax criterion, and

$$m_i = \min_j P_{ay}(A_i, S_j) = \min\{P_{ay}(A_i, S_1), P_{ay}(A_i, S_2), \ldots, P_{ay}(A_i, S_n)\} \quad \longrightarrow \quad A^*_{\text{Maximin}} = \arg\max_i m_i$$

for the Maximin criterion.

2 General 2x2 Problem

Consider the following payoff table:

Alternative	S_1	S_2
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$

Determine the decision using both the Maximax and Maximin criteria.

2.1 Solution

Step 1: Identify the alternatives and states of nature

Alternatives: A_1, A_2 States of nature: S_1, S_2.

Step 2: Calculate the Maximax values

For each alternative, identify its highest possible payoff:

$$M_1 = \max\{P_{ay}(A_1, S_1), P_{ay}(A_1, S_2)\},$$

$$M_2 = \max\{P_{ay}(A_2, S_1), P_{ay}(A_2, S_2)\}.$$

Compare the two maximum values:

$$A^*_{\text{Maximax}} = \arg\max_i M_i.$$

Thus, choose the alternative with the highest possible payoff.

Step 3: Calculate the Maximin values

For each alternative, identify its lowest possible payoff:

$$m_1 = \min\{P_{ay}(A_1, S_1), P_{ay}(A_1, S_2)\},$$

$$m_2 = \min\{P_{ay}(A_2, S_1), P_{ay}(A_2, S_2)\}.$$

Compare the two minimum values:

$$A^*_{\text{Maximin}} = \arg\max_i m_i.$$

Thus, choose the alternative with the highest worst-case payoff.

Extended payoff table

Alternative	S_1	S_2	Max	Min
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$\max\{P_{ay}(A_1, S_1), P_{ay}(A_1, S_2)\}$	$\min\{P_{ay}(A_1, S_1), P_{ay}(A_1, S_2)\}$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$\max\{P_{ay}(A_2, S_1), P_{ay}(A_2, S_2)\}$	$\min\{P_{ay}(A_2, S_1), P_{ay}(A_2, S_2)\}$

3 General 4x3 Problem

Consider the following payoff table:

Alternative	S_1	S_2	S_3
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$P_{ay}(A_1, S_3)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$P_{ay}(A_2, S_3)$
A_3	$P_{ay}(A_3, S_1)$	$P_{ay}(A_3, S_2)$	$P_{ay}(A_3, S_3)$
A_4	$P_{ay}(A_4, S_1)$	$P_{ay}(A_4, S_2)$	$P_{ay}(A_4, S_3)$

Determine the decision using both the Maximax and Maximin criteria.

3.1 Solution

Step 1: Identify the alternatives and states of nature

Alternatives: A_1, A_2, A_3, A_4 States of nature: S_1, S_2, S_3.

Step 2: Calculate the Maximax values

For each alternative, identify its highest possible payoff:

$$M_1 = \max\{P_{ay}(A_1, S_1), P_{ay}(A_1, S_2), P_{ay}(A_1, S_3)\},$$

$$M_2 = \max\{P_{ay}(A_2, S_1), P_{ay}(A_2, S_2), P_{ay}(A_2, S_3)\},$$

$$M_3 = \max\{P_{ay}(A_3, S_1), P_{ay}(A_3, S_2), P_{ay}(A_3, S_3)\},$$

$$M_4 = \max\{P_{ay}(A_4, S_1), P_{ay}(A_4, S_2), P_{ay}(A_4, S_3)\}.$$

Compare the four maximum values:

$$A^*_{\text{Maximax}} = \arg\max_i M_i.$$

Thus, choose the alternative with the highest possible payoff.

Step 3: Calculate the Maximin values

For each alternative, identify its lowest possible payoff:

$$m_1 = \min\{P_{ay}(A_1, S_1), P_{ay}(A_1, S_2), P_{ay}(A_1, S_3)\},$$

$$m_2 = \min\{P_{ay}(A_2, S_1), P_{ay}(A_2, S_2), P_{ay}(A_2, S_3)\},$$

$$m_3 = \min\{P_{ay}(A_3, S_1), P_{ay}(A_3, S_2), P_{ay}(A_3, S_3)\},$$

$$m_4 = \min\{P_{ay}(A_4, S_1), P_{ay}(A_4, S_2), P_{ay}(A_4, S_3)\}.$$

Compare the four minimum values:

$$A^*_{\text{Maximin}} = \arg\max_i m_i.$$

Thus, choose the alternative with the highest worst-case payoff.

Extended payoff table

Alternative	S_1	S_2	S_3	Max	Min
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$	$P_{ay}(A_1, S_3)$	$\max\{P_{ay}(A_1, S_1), P_{ay}(A_1, S_2), P_{ay}(A_1, S_3)\}$	$\min\{P_{ay}(A_1, S_1), P_{ay}(A_1, S_2), P_{ay}(A_1, S_3)\}$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$	$P_{ay}(A_2, S_3)$	$\max\{P_{ay}(A_2, S_1), P_{ay}(A_2, S_2), P_{ay}(A_2, S_3)\}$	$\min\{P_{ay}(A_2, S_1), P_{ay}(A_2, S_2), P_{ay}(A_2, S_3)\}$
A_3	$P_{ay}(A_3, S_1)$	$P_{ay}(A_3, S_2)$	$P_{ay}(A_3, S_3)$	$\max\{P_{ay}(A_3, S_1), P_{ay}(A_3, S_2), P_{ay}(A_3, S_3)\}$	$\min\{P_{ay}(A_3, S_1), P_{ay}(A_3, S_2), P_{ay}(A_3, S_3)\}$
A_4	$P_{ay}(A_4, S_1)$	$P_{ay}(A_4, S_2)$	$P_{ay}(A_4, S_3)$	$\max\{P_{ay}(A_4, S_1), P_{ay}(A_4, S_2), P_{ay}(A_4, S_3)\}$	$\min\{P_{ay}(A_4, S_1), P_{ay}(A_4, S_2), P_{ay}(A_4, S_3)\}$

4 Comparing the Two Criteria

The two criteria use the same payoff table but emphasize different aspects of the decision.

	Maximax	**Maximin**
Focus	Best possible payoff	Worst possible payoff
Decision rule	Choose the alternative with the highest possible payoff	Choose the alternative with the highest worst-case payoff
Attitude	Optimistic	Conservative
Risk	Accepts greater uncertainty in pursuit of a high payoff	Places greater emphasis on protection against poor outcomes
Use when	The decision-maker focuses on favorable possibilities	The decision-maker wants protection against unfavorable possibilities

A decision-maker using **Maximax** asks:

"What is the best that could happen?"

A decision-maker using **Maximin** asks:

"What is the worst that could happen, and which alternative handles it best?"

5 Contextualized 2x2 Problem - Coffee shop

A small coffee shop is considering whether to introduce a new dessert line or keep its current menu. The manager knows that the level of customer traffic during the promotional weekend is uncertain and has no reliable information about the probability of high or low traffic.

What would a perfectly logical and optimistic decision-maker choose? What would a perfectly logical and pessimistic decision-maker choose?

Alternative	High traffic (S_1)	Low traffic (S_2)
Launch dessert line (A_1)	42	8
Keep current menu (A_2)	28	20

All payoffs are expressed in thousands of dollars.

Determine the decision using both the Maximax and Maximin criteria.

5.1 Solution

Step 1: Identify the alternatives and states of nature

Alternatives: A_1 = Launch dessert line, A_2 = Keep current menu States of nature: S_1 = High traffic, S_2 = Low traffic.

Step 2: Calculate the Maximax values

For each alternative, identify its highest possible payoff:

$$M_1 = \max\{42, 8\} = 42,$$

$$M_2 = \max\{28, 20\} = 28.$$

Compare the two maximum values:

$$A^*_{\text{Maximax}} = \arg\max_i M_i = A_1.$$

Thus, the Maximax criterion recommends launching the dessert line because it offers the highest possible payoff.

Step 3: Calculate the Maximin values

For each alternative, identify its lowest possible payoff:

$$m_1 = \min\{42, 8\} = 8,$$

$$m_2 = \min\{28, 20\} = 20.$$

Compare the two minimum values:

$$A^*_{\text{Maximin}} = \arg\max_i m_i = A_2.$$

Thus, the Maximin criterion recommends keeping the current menu because it provides the highest worst-case payoff.

Extended payoff table

Alternative	S_1	S_2	Max	Min
A_1: Launch dessert line	42	8	$\max\{42, 8\} = 42$	$\min\{42, 8\} = 8$
A_2: Keep current menu	28	20	$\max\{28, 20\} = 28$	$\min\{28, 20\} = 20$

6 Contextualized 2x2 Problem - Clothing store

A local clothing store is deciding whether to launch a new advertising campaign before the holiday season or continue with its current marketing strategy. The manager knows that customer response to the campaign is uncertain and has no reliable information about the probability of strong or weak sales.

What would a perfectly logical and optimistic decision-maker choose? What would a perfectly logical and pessimistic decision-maker choose?

Alternative	Strong sales (S_1)	Weak sales (S_2)
Launch new campaign (A_1)	55	5
Keep current strategy (A_2)	35	22

All payoffs are expressed in thousands of dollars.

Determine the decision using both the Maximax and Maximin criteria.

6.1 Solution

Step 1: Identify the alternatives and states of nature

Alternatives: A_1 = Launch new campaign, A_2 = Keep current strategy States of nature: S_1 = Strong sales, S_2 = Weak sales.

Step 2: Calculate the Maximax values

For each alternative, identify its highest possible payoff:

$$M_1 = \max\{55, 5\} = 55,$$

$$M_2 = \max\{35, 22\} = 35.$$

Compare the two maximum values:

$$A^*_{\text{Maximax}} = \arg\max_i M_i = A_1.$$

Thus, the Maximax criterion recommends launching the new campaign because it offers the highest possible payoff.

Step 3: Calculate the Maximin values

For each alternative, identify its lowest possible payoff:

$$m_1 = \min\{55, 5\} = 5,$$

$$m_2 = \min\{35, 22\} = 22.$$

Compare the two minimum values:

$$A^*_{\text{Maximin}} = \arg\max_i m_i = A_2.$$

Thus, the Maximin criterion recommends keeping the current strategy because it provides the highest worst-case payoff.

Extended payoff table

Alternative	S_1	S_2	Max	Min
A_1: Launch new campaign	55	5	$\max\{55, 5\} = 55$	$\min\{55, 5\} = 5$
A_2: Keep current strategy	35	22	$\max\{35, 22\} = 35$	$\min\{35, 22\} = 22$

7 Contextualized 4x3 Problem - Technology company

A technology company is deciding which product strategy to adopt for the launch of a new device. The manager knows that market conditions are uncertain and has no reliable information about the probability of each possible level of demand. The company is considering four possible strategies: launching a premium version, launching a standard version, launching a budget version, or delaying the launch.

The level of market demand could be high, medium, or low. The estimated profit for each strategy under each possible market condition is shown in the table below.

What would a perfectly logical and optimistic decision-maker choose? What would a perfectly logical and pessimistic decision-maker choose?

Alternative	High demand (S_1)	Medium demand (S_2)	Low demand (S_3)
Premium version (A_1)	80	35	-20
Standard version (A_2)	60	42	18
Budget version (A_3)	40	32	25
Delay launch (A_4)	15	15	15

All payoffs are expressed in thousands of dollars.

Determine the decision using both the Maximax and Maximin criteria.

7.1 Solution

Step 1: Identify the alternatives and states of nature

Alternatives: A_1 = Premium version, A_2 = Standard version, A_3 = Budget version, A_4 = Delay launch

States of nature: S_1 = High demand, S_2 = Medium demand, S_3 = Low demand.

Step 2: Calculate the Maximax values

For each alternative, identify its highest possible payoff:

$$M_1 = \max\{80, 35, -20\} = 80,$$

$$M_2 = \max\{60, 42, 18\} = 60,$$

$$M_3 = \max\{40, 32, 25\} = 40,$$

$$M_4 = \max\{15, 15, 15\} = 15.$$

Compare the four maximum values:

$$A^*_{\text{Maximax}} = \arg\max_i M_i = A_1.$$

Thus, the Maximax criterion recommends launching the premium version because it offers the highest possible payoff.

Step 3: Calculate the Maximin values

For each alternative, identify its lowest possible payoff:

$$m_1 = \min\{80, 35, -20\} = -20,$$

$$m_2 = \min\{60, 42, 18\} = 18,$$

$$m_3 = \min\{40, 32, 25\} = 25,$$

$$m_4 = \min\{15, 15, 15\} = 15.$$

Compare the four minimum values:

$$A^*_{\text{Maximin}} = \arg\max_i m_i = A_3.$$

Thus, the Maximin criterion recommends launching the budget version because it provides the highest worst-case payoff.

Extended payoff table

Alternative	S_1	S_2	S_3	Max	Min
A_1: Premium version	80	35	-20	$\max\{80, 35, -20\} = 80$	$\min\{80, 35, -20\} = -20$
A_2: Standard version	60	42	18	$\max\{60, 42, 18\} = 60$	$\min\{60, 42, 18\} = 18$
A_3: Budget version	40	32	25	$\max\{40, 32, 25\} = 40$	$\min\{40, 32, 25\} = 25$
A_4: Delay launch	15	15	15	$\max\{15, 15, 15\} = 15$	$\min\{15, 15, 15\} = 15$